

Project Brief: Decision-grade Drought Risk Brief / Dashboard

Owner: Sophia Mbataru

Users: Drought desk + NGO program teams + field ops

Target geography: East and Central Africa

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1. Problem & Urgency

Climate extremes are intensifying, and much of East and Central Africa is experiencing prolonged drought punctuated by erratic rainfall. County planners, humanitarian actors, and financial institutions often lack timely, interpretable data to act early and allocate resources effectively. To build resilience, we need transparent tools that translate raw climate signals into practical drought-risk insights.

2. Primary Users

- County-level planners and disaster risk managers
- Government agencies (e.g., agriculture, water, early warning units)
- Agricultural lenders and microfinance institutions
- Non-technical users (NGO staff, extension officers, program managers)

3. Decisions Supported

The dashboard supports:

- Early-season hotspot detection
- Water resource planning and drought response prioritisation
- Risk-scoring for drought-linked financing or safety nets
- Monitoring climate stress for resilience program design

4. Deliverable

A public, interactive dashboard that transforms satellite data (CHIRPS rainfall, MODIS NDVI, SMAP soil moisture) into a harmonised **Drought Risk Index (0–100)**. Key features:

- Calendar-adjusted monthly climatologies
- Quantile-scaled choropleth maps
- Admin-level summaries (national to county)
- MoM/YoY trends and downloadable tables
- Mobile-friendly UI for low-bandwidth settings

5. Success Metrics (Pilot-Ready)

- 80%+ match with independent drought bulletins (e.g. NDMA, FEWS NET)
- Time-to-insight reduced by >60% vs manual workflows
- Usability scores from >3 non-technical test users
- Map load times <5 seconds in low-connectivity settings

6. Scope

In-scope: Kenya, Uganda, Somalia, Ethiopia, South Sudan, Rwanda, plus select Middle East regions

Out-of-scope (for now): Automated forecasts, crop yield projections

7. Assumptions & Dependencies

- Continued access to open satellite datasets (e.g., CHIRPS, MODIS, SMAP via GEE)
- Stable data storage and hosting environment
- Feedback from at least one institutional pilot partner

8. Timeline & Risks

Timeline: 4–6 weeks

- Week 1–2: Data harmonisation + index design
- Week 3–4: UI/dashboard build + internal testing
- Week 5–6: User testing + deployment

Risks: Satellite data latency, mapping at too-coarse resolution, misalignment with existing planning cycles

9. Definition of Done

The dashboard is complete when it:

- Accurately flags drought-affected regions in line with trusted sources >75% of the time
- Is deployed publicly with documentation
- Has at least one feedback session from a target user group

